Package ‘dynTools’

June 13, 2026

Title Dynamic Modeling Utilities

Version 0.0.0.9000

Description Utility functions for data preparation when fitting dynamic models.

URL https://github.com/jeksterslab/dynTools,
https://jeksterslab.github.io/dynTools/

BugReports https://github.com/jeksterslab/dynTools/issues

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Encoding UTF-8

Roxygen list(markdown = TRUE)

VignetteBuilder knitr

Depends R (>= 4.1.0)

Suggests knitr, rmarkdown, testthat

Config/roxygen2/version 8.0.0.9000

NeedsCompilation no

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Contents

`CheckDynData` *Check Dynamic Modeling Data*

Description

The function checks whether a data frame has the variables and structure required by the dynamic modeling utility functions.

Usage

```
CheckDynData(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  require_unique = TRUE,
  require_numeric_time = TRUE,
  require_numeric_observed = TRUE,
  require_numeric_covariates = FALSE,
  min_rows = 2L
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

`require_unique` Logical. If `TRUE`, require unique `id`-`time` combinations.

`require_numeric_time`
: Logical. If `TRUE`, require the time variable to be numeric.

`require_numeric_observed`
: Logical. If `TRUE`, require observed variables to be numeric.

`require_numeric_covariates`
: Logical. If `TRUE`, require covariates to be numeric.

`min_rows` Positive integer. Minimum number of rows required per ID.

Value

Returns `TRUE` invisibly if all checks pass.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
CheckDynData(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

DeleteInitialNA *Delete for NAs in Initial Row By ID*

Description

The function removes initial rows by ID if they contain missing values. This process is repeated until the first row per ID no longer has missing observations.

Usage

```
DeleteInitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

DeleteInitialNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

DeltaTByID *Summarize Time Intervals by ID*

Description

The function returns a diagnostic table describing the spacing of the time variable within each ID.

Usage

```
DeltaTByID(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  tolerance = sqrt(.Machine$double.eps)
)
```

Arguments

data	Data frame. A data frame object.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. Optional vector of character strings of the names of the observed variables in the data.
covariates	Character vector. Optional vector of character strings of the names of the covariates in the data.
tolerance	Numeric. Tolerance used to determine whether the time intervals are regular.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = c(1, 2, 3, 1, 3, 4),
  y = rnorm(6)
)
data

DeltaTByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y"
)
```

DetrendByID *Detrend Observed Variables by ID*

Description

The function removes polynomial time trends from observed variables within each ID. Missing observed values are ignored when estimating the trend and remain missing in the detrended variables.

Usage

```
DetrendByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  degree = 1L,
  replace = FALSE,
  keep_mean = TRUE,
  prefix = "detrend"
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.

observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
degree	Non-negative integer. Degree of the polynomial trend. Use `degree = 0` for mean centering by ID when `keep_mean = FALSE`. If `degree = 0` and `keep_mean = TRUE`, the original observed values are returned.
replace	Logical. If `TRUE`, replace observed variables with detrended variables. If `FALSE`, append detrended variables to the data frame.
keep_mean	Logical. If `TRUE`, preserve the original within-ID mean after detrending. If `FALSE`, return ordinary residuals from the within-ID trend model.
prefix	Character string. Prefix for detrended variables when `replace = FALSE`.

Details

If `keep_mean = TRUE`, the within-ID observed mean is added back after removing the fitted time trend. This removes temporal drift while preserving each ID's observed-variable level. If `keep_mean = FALSE`, the output is the ordinary residual from the within-ID polynomial trend model.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y = rep(1:4, times = 2)
)
data

DetrendByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  degree = 1,
```

```
  keep_mean = TRUE
)

DetrendByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  degree = 1,
  keep_mean = FALSE
)
```

ElapsedTimeByID *Create Elapsed Time by ID*

Description

The function creates an elapsed-time variable within ID. For date-time variables, elapsed time is computed with `difftime()`. For numeric time variables, elapsed time is computed by subtraction, optionally converting from `input_units` to `units`.

Usage

```
ElapsedTimeByID(
  data,
  id,
  time,
  output = "time_elapsed",
  units = "days",
  origin = c("by_id", "global"),
  input_units = NULL,
  replace = FALSE
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`output`	Character string. Name of the elapsed-time variable.
`units`	Character string. Units for the output elapsed time.

`origin`	Character string. If `origin = "by_id"`, each ID's origin is its own minimum non-missing time. If `origin = "global"`, all IDs use the global minimum non-missing time.
`input_units`	Character string or `NULL`. Units of numeric input time. If `NULL`, numeric time is assumed to already be in the desired output scale and only subtraction is performed.
`replace`	Logical. If `TRUE`, replace the original `time` variable. If `FALSE`, append `output`.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-02 00:00:00",
      "2020-01-03 12:00:00",
      "2020-01-04 00:00:00"
    ),
    tz = "UTC"
  )
)
data

ElapsedTimeByID(
  data = data,
  id = "id",
  time = "datetime",
  output = "time",
  units = "days"
)
```

`FilterByID` *Filter Dynamic Modeling Data by ID*

Description

The function removes IDs that do not satisfy minimum data requirements.

Usage

```
FilterByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_rows = 2L,
  min_complete = 2L,
  max_prop_missing = 1,
  allow_initial_na = TRUE,
  allow_all_missing = FALSE
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`min_rows`	Non-negative integer. Minimum number of rows required per ID.
`min_complete`	Non-negative integer. Minimum number of complete observed rows required per ID.
`max_prop_missing`	Numeric. Maximum allowed proportion of missing values across observed variables.
`allow_initial_na`	Logical. If `FALSE`, remove IDs where the initial row contains missing values.
`allow_all_missing`	Logical. If `FALSE`, remove IDs where all observed values are missing.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = rep(1:3, each = 3),
  time = rep(1:3, times = 3),
  y = c(1, 2, 3, NA, NA, 4, NA, NA, NA)
)
data

FilterByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  min_complete = 2
)
```

`InitialNA` *Check for NAs in Initial Row By ID*

Description

The function checks if there are missing values for the initial row by ID.

Usage

```
InitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a vector of ID numbers where the initial row has any missing value.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

InitialNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

InsertNA *Insert NAs for Missing Observations*

Description

The function creates a sequence of time values. It starts with the smallest time value as the starting point and the largest time value as the endpoint. The sequence is incremented by `delta_t`. This new sequence is combined with the existing empirical time values. For any specific time value where there are no observations, NAs are inserted.

Usage

```
InsertNA(data, id, time, observed, covariates = NULL, delta_t)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`delta_t`	Positive number. Time interval.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 4, 1, 3, 4),
  y1 = c(10, 11, 13, 20, 22, 23),
  y2 = c(5, 6, 8, 15, 17, 18)
)
data

InsertNA(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  delta_t = 1
)
```

LagByID *Create Lagged Variables by ID*

Description

The function creates lagged observed variables within ID. Lagged values never cross ID boundaries.

Usage

```
LagByID(data, id, time, observed, covariates = NULL, lags = 1L, prefix = "lag")
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.
`lags`	Positive integer vector. Lags to create.
`prefix`	Character string. Prefix for the lagged variable names.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
data

LagByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  lags = 1
)
```

MakeClockTime *Make Clock Time*

Description

The function combines a date and a clock time into a POSIXct vector. Clock time can be supplied as decimal hours, for example 13.5, or as a character time, for example "13:30" or "13:30:00".

Usage

```
MakeClockTime(
  date,
  time,
  tz = "UTC",
  date_formats = c("%m/%d/%y", "%m/%d/%Y", "%Y-%m-%d"),
  invalid = c("NA", "error")
)
```

Arguments

`date`	Date vector. Dates can be supplied as `Date`, `POSIXt`, or character values.
`time`	Numeric or character vector. Clock time in decimal hours, `"HH:MM"`, or `"HH:MM:SS"` format.
`tz`	Character string. Time zone passed to `as.POSIXct()`.
`date_formats`	Character vector. Date formats used to parse character dates.
`invalid`	Character string. If `invalid = "NA"`, invalid clock times are returned as `NA`. If `invalid = "error"`, invalid clock times throw an error.

Value

Returns a `POSIXct` vector.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
MakeClockTime(
  date = "2020-01-02",
  time = 13.5,
  tz = "America/New_York"
)

MakeClockTime(
  date = "01/02/20",
  time = "13:30",
  tz = "America/New_York"
)
```

`PlotByID` *Plot Observed Variables by ID*

Description

The function creates one time-series plot for each observed variable. Within each plot, trajectories are overlaid by ID.

Usage

```
PlotByID(
  data,
  id,
  time,
  observed,
  ids = NULL,
  times = NULL,
  type = "b",
  pch = NULL,
  lty = NULL,
  col = NULL,
  xlab = NULL,
  ylab = NULL,
  main = NULL,
  xlim = NULL,
  ylim = NULL,
  legend = FALSE,
  ask = NULL,
  ...
)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`ids`	Optional vector. ID values to plot. If `NULL`, all available IDs are plotted.
`times`	Optional vector of time values. If `NULL`, all available time points are plotted. If supplied, the range of `times` is used to subset the data. For example, `times = c(0, 9)` plots all rows with time values from 0 to 9, inclusive.
`type`	Character string. Plot type passed to `graphics::lines()`. Default is `"b"`.
`pch`	Optional vector of plotting characters. If `NULL`, plotting characters are generated automatically by ID. Recycled to match the number of plotted IDs.
`lty`	Optional vector of line types. If `NULL`, line types are generated automatically by ID. Recycled to match the number of plotted IDs.
`col`	Optional vector of colors. If `NULL`, colors are generated automatically by ID. Recycled to match the number of plotted IDs.
`xlab`	Optional character string. Label for the x-axis. If `NULL`, the name of the time variable is used.

ylab	Optional character string. Label for the y-axis. If NULL, the observed variable name is used.
main	Optional character string. Plot title prefix. If NULL, the observed variable name is used.
xlim	Optional vector of length 2. Limits for the x-axis.
ylim	Optional numeric vector of length 2. Limits for the y-axis. If NULL, limits are computed separately for each observed variable.
legend	Logical. If TRUE, add a legend identifying IDs.
ask	Logical. If TRUE, ask before drawing the next observed-variable plot. The default is TRUE in interactive sessions when plotting more than one observed variable.
...	Additional arguments passed to graphics::lines().

Value

Invisibly returns the IDs that were plotted.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

Examples

```
data <- data.frame(
  id = rep(1:4, each = 5),
  time = rep(1:5, times = 4),
  y1 = c(
    1, 2, 3, 4, 5,
    2, 3, 4, 5, 6,
    3, 4, 5, 6, 7,
    4, 5, 6, 7, 8
  ),
  y2 = c(
    5, 4, 3, 2, 1,
    6, 5, 4, 3, 2,
    7, 6, 5, 4, 3,
    8, 7, 6, 5, 4
  ),
  y3 = c(
    1, 1, 2, 2, 3,
    2, 2, 3, 3, 4,
    3, 3, 4, 4, 5,
```

```
    4, 4, 5, 5, 6
  )
)
data

PlotByID(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3),
  ask = FALSE
)

PlotByID(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3),
  ids = 1:3,
  times = c(1, 3),
  legend = TRUE,
  ask = FALSE
)
```

`RegularizeTimeByID` *Regularize Time by ID*

Description

The function inserts rows with missing values so that time follows a regular grid. A global grid uses the same time sequence for every ID. An ID-specific grid uses each ID's own minimum and maximum observed time values. If `method = "preserve"`, the function completes the regular grid while retaining empirical off-grid time values. If `method = "snap"`, empirical time values are snapped to the nearest grid point and the returned data use only grid time values.

Usage

```
RegularizeTimeByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  delta_t,
  grid = c("global", "by_id"),
  method = c("preserve", "snap")
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
delta_t	Positive number. Time interval.
grid	Character string. If `grid = "global"`, use one time grid from the global minimum to the global maximum time value. If `grid = "by_id"`, use an ID-specific time grid from each ID's minimum to maximum time value.
method	Character string. If `method = "preserve"`, complete the regular grid but preserve observed off-grid time values. If `method = "snap"`, snap observed time values to the nearest grid point so the output contains only regular grid times. When multiple rows for the same ID snap to the same grid point, the row closest to the grid point is kept; ties are broken by the largest number of non-missing observed/covariate values and then by original order.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  time = c(1, 3, 1, 2),
  y = c(10, 12, 20, 21)
)
data
```

```
RegularizeTimeByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 1,
  grid = "by_id"
)

RegularizeTimeByID(
  data = data.frame(
    id = c(1, 1),
    time = c(1.0, 1.7),
    y = c(10, 17)
  ),
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 0.5,
  grid = "by_id",
  method = "snap"
)
```

`ReplaceMissingCode` *Replace Missing-Value Codes*

Description

The function replaces user-specified missing-value codes with `NA` in a data frame. The replacement is applied column by column and preserves the original column classes whenever possible.

Usage

```
ReplaceMissingCode(data, values = c(-999, "-999"), columns = names(data))
```

Arguments

`data`	Data frame.
`values`	Vector. Values to replace with `NA`.
`columns`	Character vector. Names of the columns where replacement should be applied. If `NULL`, no columns are modified.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

Examples

```
data <- data.frame(
  id = 1:3,
  y = c(1, -999, 3),
  x = c("a", "-999", "c"),
  stringsAsFactors = FALSE
)
data

ReplaceMissingCode(
  data = data,
  values = c(-999, "-999")
)
```

ResolveDuplicateIDTime

Resolve Duplicate ID-Time Rows

Description

The function resolves exact duplicate `id-time` rows using a simple deterministic rule. This is useful before inserting missing rows on a time grid because grid-completion functions generally require unique `id-time` combinations.

Usage

```
ResolveDuplicateIDTime(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  method = c("first", "last", "max_complete"),
  order_by = NULL,
  decreasing = FALSE
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
method	Character string. Rule used to choose one row from each duplicated `id-time` group. If `method = "first"`, keep the first row after sorting by `id` and `time`. If `method = "last"`, keep the last row after sorting by `id` and `time`. If `method = "max_complete"`, keep the row with the largest number of non-missing values in `observed`.
order_by	Character vector. Optional columns used to break ties after the main `method` rule.
decreasing	Logical. If `TRUE`, sort `order_by` columns in decreasing order.

Value

Returns a data frame with unique `id-time` rows.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1),
  time = c(0, 0, 1),
  y1 = c(NA, 1, 2),
  y2 = c(NA, 1, 3)
)
data

ResolveDuplicateIDTime(
```

```
    data = data,
    id = "id",
    time = "time",
    observed = c("y1", "y2"),
    method = "max_complete"
  )
```

RoundClockTime *Round Clock Time*

Description

The function rounds, floors, or ceilings clock times to a specified unit. It supports `POSIXt`, `Date`, and numeric time values.

Usage

```
RoundClockTime(
  x,
  unit = "hour",
  method = c("round", "floor", "ceiling"),
  origin = "1970-01-01",
  tz = "UTC"
)
```

Arguments

`x`	A `POSIXt`, `Date`, or numeric vector.
`unit`	Character string. Unit to round to. Supported values are `"second"`, `"minute"`, `"hour"`, `"day"`, and their plural forms.
`method`	Character string. One of `"round"`, `"floor"`, or `"ceiling"`.
`origin`	Origin for numeric time values. Only used if `x` is numeric.
`tz`	Character string. Time zone used when numeric time values are converted to `POSIXct`.

Value

Returns a vector with rounded times.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), ScaleByID(), SubsetByID(), SummaryByID()

Examples

```
x <- as.POSIXct(
  "2020-01-01 10:31:00",
  tz = "America/New_York"
)

RoundClockTime(
  x = x,
  unit = "hour"
)
```

ScaleByID *Scale by ID*

Description

The function scales the data by ID.

Usage

```
ScaleByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  scale = TRUE,
  obs_skip = NULL,
  cov_skip = NULL
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.

time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
scale	Logical. If `scale = TRUE`, standardize by `id`. If `scale = FALSE`, mean center by `id`.
obs_skip	Character vector. A vector of character strings of the names of the observed variables to skip centering/scaling.
cov_skip	Character vector. A vector of character strings of the names of the covariates to skip centering/scaling.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `SubsetByID()`, `SummaryByID()`

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 22, 24, 26
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    26, 24, 22, 20
  )
)
data

ScaleByID(
  data = data,
  id = "id",
  time = "time",
```

```
  observed = c("y1", "y2")
)

ScaleByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  scale = FALSE
)
```

SubsetByID *Subset Data Set by ID*

Description

The function creates a list of data frames for each ID.

Usage

```
SubsetByID(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a list by ID numbers.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), PlotByID(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SummaryByID()

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 21, 22, 23
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    23, 22, 21, 20
  )
)
data

SubsetByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

SummaryByID *Summarize Dynamic Modeling Data by ID*

Description

The function returns a diagnostic table with one row per ID.

Usage

```
SummaryByID(data, id, time, observed, covariates = NULL)
```

Arguments

`data`	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.

`id`	Character string. A character string of the name of the ID variable in the data.
`time`	Character string. A character string of the name of the TIME variable in the data.
`observed`	Character vector. A vector of character strings of the names of the observed variables in the data.
`covariates`	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = c(1, NA, 3, 4, 5, 6),
  y2 = c(1, 2, 3, NA, NA, NA)
)
data

SummaryByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

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